

Digital System Design

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Smart Parking System Report

Team 4

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Project Idea

Our project is a Smart Parking System implemented using an FPGA Altera MAX 10, an SG90 servo motor, and two IR sensors. The system manages entry through a single gate and keeps track of the number of available parking spots (starting with 8). The number of empty spots is always displayed on the output.

The system supports two operating modes, selected using a mode switch:

- **Autonomous Mode**

The gate opens automatically when a car is detected by Sensor 1.

After the car passes through and triggers Sensor 2, the gate automatically closes.

The gate will open only if there are available parking spots.

- **Manual Mode**

The user manually controls the gate using a dedicated switch (open/close).

However, the gate will not open if the parking lot is full.

Additionally, a reset switch is included to restore the system to its initial state:

Resets the available parking spots to 8.

Inputs:

1. Clock
2. Reset
3. Mode_Switch
4. Sensor_In
5. Sensor_Out
6. Gate_Switch

Outputs:

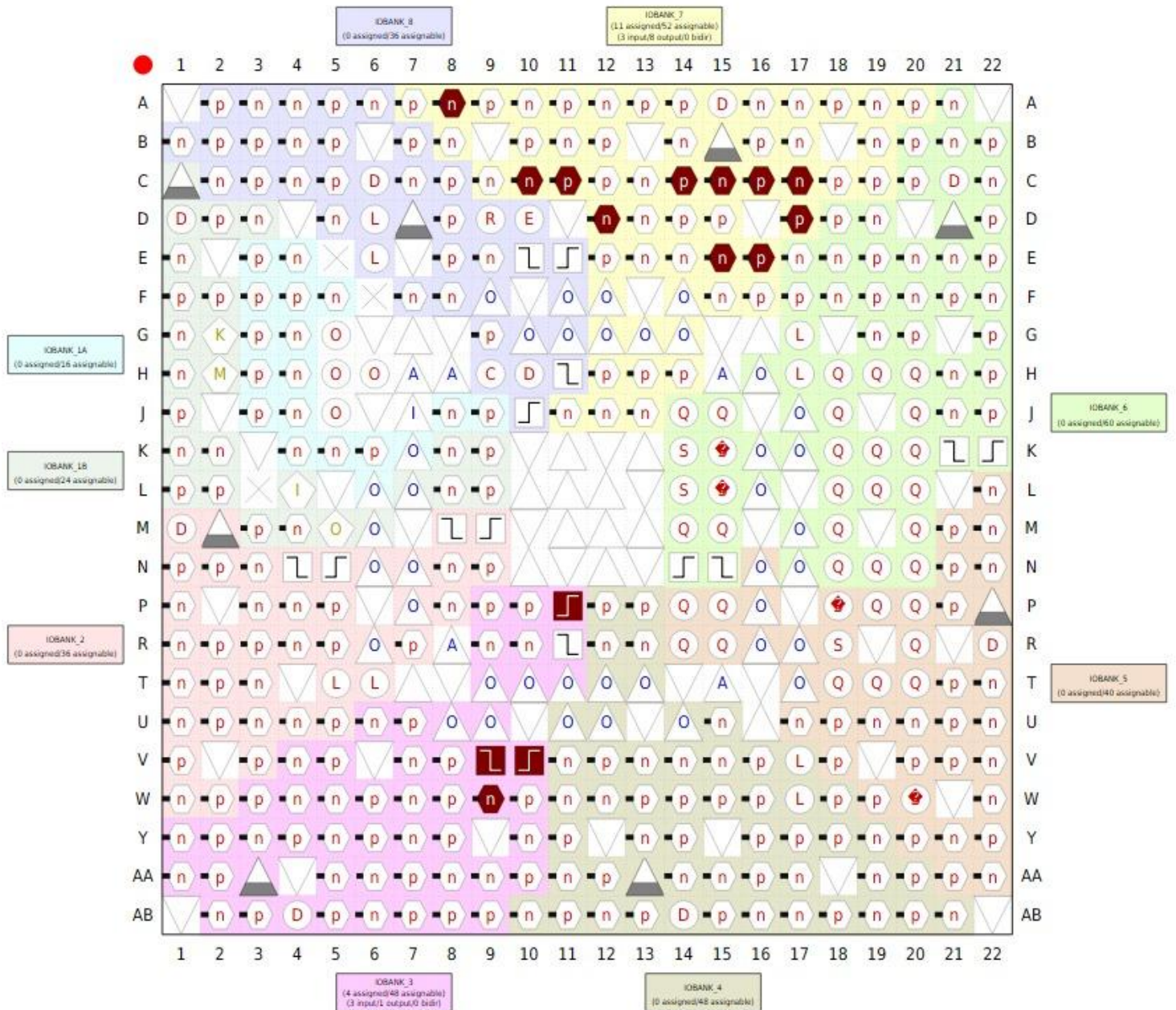
1. Final_Gate_State
2. Servo_pwm
3. Spots_Display

Pin Assignments

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Strength	Slew Rate
 spots_display[0]	Output	PIN_C14	7	B7_N0	PIN_C14	2.5 V		12mA (default)	2 (default)
 spots_display[1]	Output	PIN_E15	7	B7_N0	PIN_E15	2.5 V		12mA (default)	2 (default)
 spots_display[2]	Output	PIN_C15	7	B7_N0	PIN_C15	2.5 V		12mA (default)	2 (default)
 spots_display[3]	Output	PIN_C16	7	B7_N0	PIN_C16	2.5 V		12mA (default)	2 (default)
 spots_display[4]	Output	PIN_E16	7	B7_N0	PIN_E16	2.5 V		12mA (default)	2 (default)
 spots_display[5]	Output	PIN_D17	7	B7_N0	PIN_D17	2.5 V		12mA (default)	2 (default)
 spots_display[6]	Output	PIN_C17	7	B7_N0	PIN_C17	2.5 V		12mA (default)	2 (default)
 servo_pwm	Output	PIN_V10	3	B3_N0	PIN_V10	2.5 V		12mA (default)	2 (default)
 final_gate_state	Output	PIN_A8	7	B7_N0	PIN_A8	2.5 V		12mA (default)	2 (default)
 sensor_out	Input	PIN_W9	3	B3_N0	PIN_W9	2.5 V		12mA (default)	
 sensor_in	Input	PIN_V9	3	B3_N0	PIN_V9	2.5 V		12mA (default)	
 reset	Input	PIN_C10	7	B7_N0	PIN_C10	2.5 V		12mA (default)	
 mode_switch	Input	PIN_C11	7	B7_N0	PIN_C11	2.5 V		12mA (default)	
 gate_switch	Input	PIN_D12	7	B7_N0	PIN_D12	2.5 V		12mA (default)	
 clk	Input	PIN_P11	3	B3_N0	PIN_P11	2.5 V		12mA (default)	

Node Name	Direction	Location
 spots_display[0]	Output	PIN_C14
 spots_display[1]	Output	PIN_E15
 spots_display[2]	Output	PIN_C15
 spots_display[3]	Output	PIN_C16
 spots_display[4]	Output	PIN_E16
 spots_display[5]	Output	PIN_D17
 spots_display[6]	Output	PIN_C17
 servo_pwm	Output	PIN_V10
 final_gate_state	Output	PIN_A8
 sensor_out	Input	PIN_W9
 sensor_in	Input	PIN_V9
 reset	Input	PIN_C10
 mode_switch	Input	PIN_C11
 gate_switch	Input	PIN_D12
 clk	Input	PIN_P11

MAX 10 - 10M50DAF484C7G



Brief Explanation of the code:

This VHDL implementation of a smart parking system is synchronous and controlled by a clock and reset signal, where the functionality revolves around an integer signal 'spots' that tracks available parking spaces from 8 down to 0. The architecture uses two operational modes selected by 'mode_switch': autonomous mode has 'sensor_in' and 'sensor_out' inputs to automatically control a servo motor via precisely generated PWM signals, with constants defining a 20ms PWM period (PWM_PERIOD = 1000000 cycles for 50MHz clock), 1ms pulse width for -90° closed position (PWM_MIN = 125000), and 2ms pulse width for +90° open position (PWM_MAX = 75000), while manual mode allows direct gate control through 'gate_switch'. The system incorporates safety timing mechanisms using 'MOVEMENT_DELAY' constants (7,500,000 cycles) implemented via 'delay_timer_in' and 'delay_timer_out' counters that prevent multiple triggers and ensure complete gate movement before state transitions. A dedicated 'seg_encode' function converts the integer spot count to 7-segment display patterns using case statements with active-low configurations, while parallel processes handle the main parking logic and PWM generation separately, with the main process managing spot decrementation only when vehicles fully exit in autonomous mode or during manual closure, and the PWM process generating servo control signals through counter-comparison logic that produces the required pulse widths for precise angular positioning of the gate mechanism.

Results:

The smart parking system worked correctly. It successfully counted parking spots from 8 down to 0 and showed the number on the display. The gate opened and closed properly using both automatic mode and manual mode. The system reset correctly when needed and handled all situations without errors.